

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

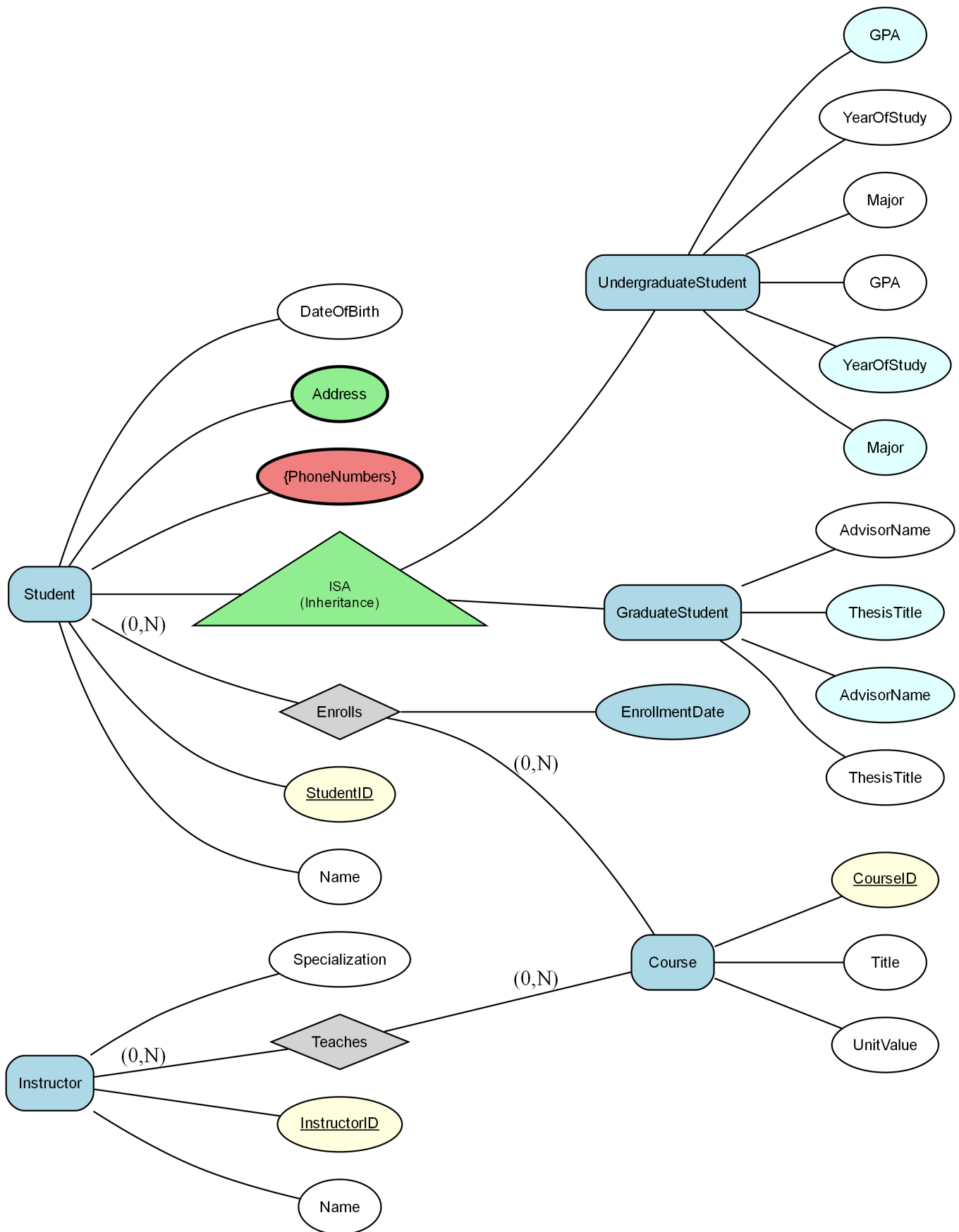
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

A university manages information about its courses, students, and instructors. Each course has a unique CourseID, a title, and a unit value. Students are assigned a unique StudentID, name, and date of birth, and can enroll in multiple courses. Instructors, identified by a unique InstructorID, have a name and specialization. Courses can be taught by multiple instructors, while each instructor can teach multiple courses. There is a many-to-many 'Teaches' relationship between Courses and Instructors, indicating which instructors are assigned to teach which courses. Additionally, students can have a multivalued attribute for PhoneNumbers and a composite attribute for Address, consisting of Street, City, and ZipCode. The university also has a specialization for graduate and undergraduate students, indicating their level of study.

Figure: EER



a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly.

(17)

b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer.

(3)

Question Q2 (15 marks)

Consider a relation R(ProductID, ProductName, SupplierID, SupplierName, Price) with the following set of functional dependencies F over R: $F = \{\{ProductID \rightarrow ProductName, SupplierID \rightarrow SupplierName, ProductID \rightarrow Price\}\}$. Analyze this relation for keys and normalization forms.

- a) Find all the keys in relation R using the attribute closure method. (4)
- b) Is R in 3NF? Give reasons for your conclusion. (3)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8)

Question Q3 (25 marks)

A university is developing a comprehensive database system to manage its courses, students, and faculty. The database system will handle complex relations such as student enrollments in courses, faculty assignments to classes, and grading systems. The administration team has outlined different roles within the system to facilitate management and access. Olivia is the head database administrator responsible for the entire database infrastructure. Liam and Zoe are junior database administrators supporting Olivia. Liam manages the courses database, while Zoe oversees the student records database. Emma is a database developer working under Liam and Zoe, responsible for designing the schema of the courses and student tables. Adam and Mia are data entry operators tasked with entering and updating course and student information. Explore the structure of this database management system through the following questions.

- a) Write SQL queries to accept or refute the above statement justifying your answer. (2)

- b) Code Segment:

```
``sql
PreparedStatement pstmt = connection.prepareStatement(
    "SELECT * FROM Employees WHERE Department = ?");
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
``
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2)

- c) Briefly explain the options available for transferring data between two sources such as between tables and servers. (2)
- d) Briefly explain how authentication and authorization are achieved in SQL Server. (3)

- e) Olivia is the head database administrator tasked with managing the overall database system access. Write a T-SQL statement to create a login for Olivia with Windows authentication. (3)
- f) Provide Olivia with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (2)
- g) Assuming Zoe's username is 'zoe.z', write T-SQL statements to assign her the responsibility of managing the student records database using a user-defined server role. (5)
- h) Assuming Emma's login name is 'emma.e', write T-SQL statements to grant her permission to create tables and databases. (3)
- i) Assuming Mia's login name is 'mia.m', write T-SQL statements to allow her to perform data entry tasks on the courses and student records databases. (3)

Question Q4 (40 marks)

A local library database stores information about members, books, and loans. The 'Members' table contains member_id, name, address, and total_fines. The 'Books' table holds book_id, title, author, and status. The 'Loans' table records loan_id, member_id, book_id, issue_date, and due_date. Utilize this structure to manage loans and fines effectively.

- a) ii. Write SQL queries to find the member who has the highest total amount of fines than all the other total fine amounts. Find the member's id and address. (7)
- b) iii. Find the book titles, authors, and member names of books currently on loan. (8)
- c) Create a function to calculate the total fine amount for a given member. This function should account for fines with a late penalty of 10% applied to those marked as 'overdue' payment status. (12)
- d)) Create a trigger that automatically updates the 'TotalFineAmount' column in the 'Members' table whenever a new fine is added, or an existing fine is updated. The 'TotalFineAmount' column should contain the total fine amount for each member. Ensure the trigger adjusts the 'TotalFineAmount' column accurately to reflect any partial payments made by members. The member pays only 50% of the total fine amount as an initial payment when the payment status is marked as 'Partial'. (13)